



Assignment 01: OOP Coding Café, Unit 01

Q. No. Question (Attempt Any 06 Questions)

- 1 Explain the need for Object-Oriented Programming (OOP). Differentiate between Procedural Programming and Object-Oriented Programming with respect to data security, reusability, abstraction, program design, and maintainability. Give a suitable example.
- 2 Define class, object, attributes, and methods in C++. Explain the relationship between a class and its objects with a suitable real-world example.
- 3 Explain static binding and dynamic binding. How does binding affect the execution of a program? Illustrate the concept with a suitable C++ example.
- 4 Define a class Student with private data members rollNo, name, and marks. Implement public member functions to accept and display the data. Demonstrate the use of access specifiers and data abstraction.
- 5 Design a C++ class BankAccount in which the class interface is separated from its implementation. Write the class declaration with function prototypes and define the member functions separately using the scope resolution operator ::.
- 6 What is a function prototype? Explain its purpose and syntax. Write a C++ program using a function prototype to calculate the factorial and Fibonacci series using separate functions.
- 7 Create a class Rectangle with data members length and breadth. Implement default, parameterized, and copy constructors and a destructor. Demonstrate the execution order of these constructors and destructor.
- 8 Explain the memory requirements of objects in C++. Write a program containing a class with data members and member functions and discuss which components consume memory for each object and which are shared.
- 9 Create a class Employee with a static data member employeeCount that keeps track of the number of objects created. Implement a static member function to display the total number of employees.
- 10 Write a C++ program to demonstrate inline functions and friend functions. Create a class Distance with private members representing feet and inches. Use a friend function to add two Distance objects and display the result.
- 11 Design two classes, Student and Result, where a friend class Result accesses the private data of Student to calculate and display the student's total, average, and grade. Explain why a friend class is used and discuss its advantages and limitations.

Assignment 02,OOP, Coding Café Unit 02

- Q. No. Question (Attempt Any 06 Questions)
- 1 Explain the concept of encapsulation in C++. How do access modifiers (private, protected, and public) help achieve encapsulation? Give a suitable example.
 - 2 Write a C++ program to declare, initialize, and access a variable using a pointer. Demonstrate the use of the address-of (&) and indirection (*) operators.
 - 3 Write a C++ program to dynamically allocate memory for an integer array using new. Accept values from the user, display them, and release the allocated memory using delete[]. Explain why proper memory deallocation is necessary.
 - 4 Create a class Employee containing private data members id and salary. Create an object dynamically using the new operator and access its members through a pointer to an object. Release the object using delete.
 - 5 Explain the purpose of the this pointer in C++. Write a program in which the this pointer is used to distinguish between class data members and function parameters having the same name.
 - 6 Write a C++ program demonstrating an array of pointers. Store the addresses of five integers in an array of pointers and display the values using these pointers.
 - 7 Explain what a function pointer is. Write a C++ program using a function pointer to perform arithmetic operations such as addition, subtraction, multiplication, and division.
 - 8 Write a C++ program demonstrating a pointer to pointer (**). Create an integer variable, a pointer to that variable, and a pointer to the first pointer. Display the value using all levels of indirection.
 - 9 Write a C++ program demonstrating passing pointers to functions. Create a function that accepts the address of two integers and modifies their values.
 - 10 Differentiate between call by value and call by reference. Write a C++ program to swap two numbers using both techniques and explain why the changes made by call by value do not affect the original variables.
 - 11 Write a C++ program that demonstrates null pointer and void pointer. Initialize a pointer to nullptr, safely test it before dereferencing, and use a void* pointer to access an integer and a floating-point value using appropriate type casting.